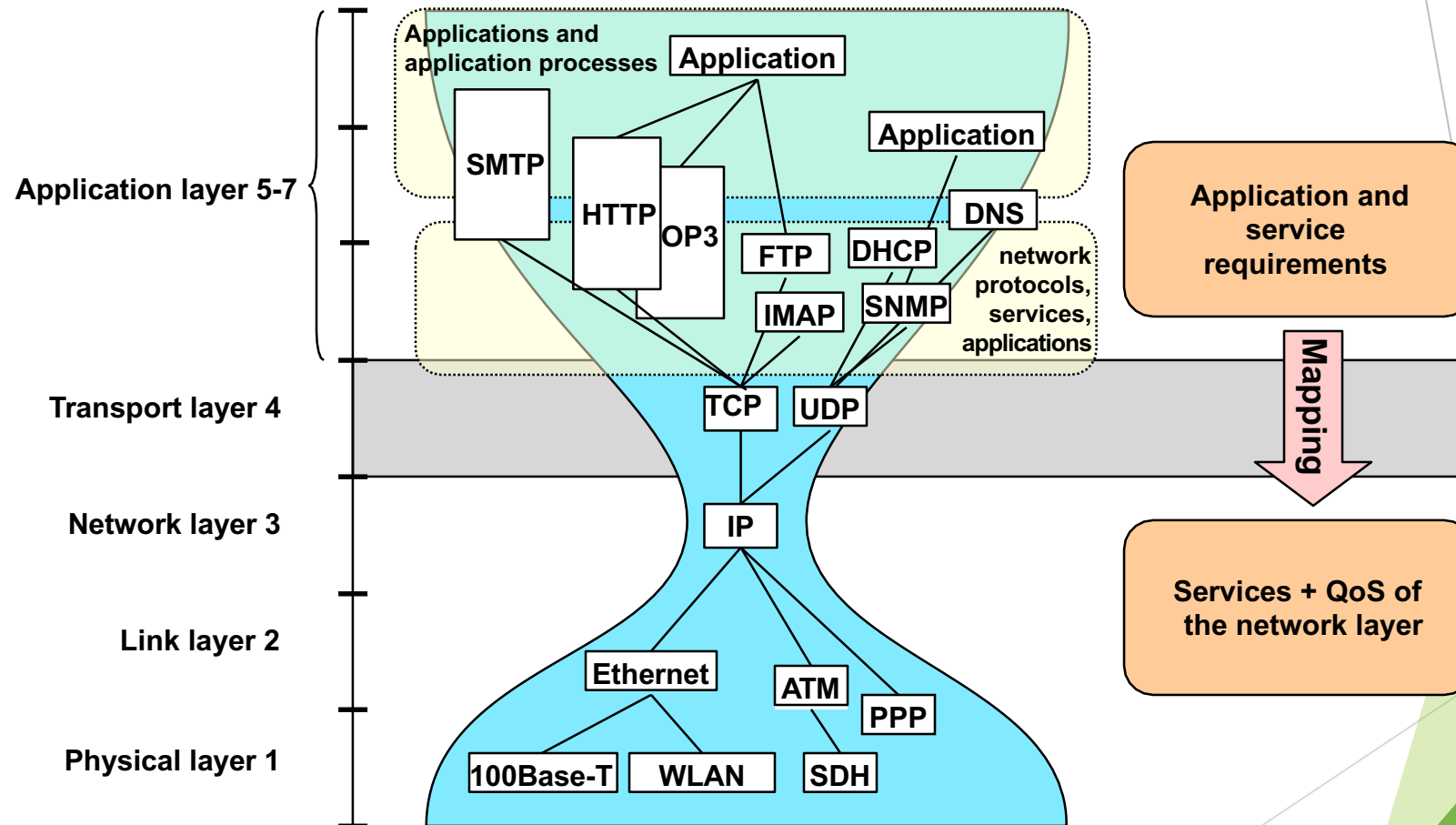


UDP & TCP Basics (Part I)

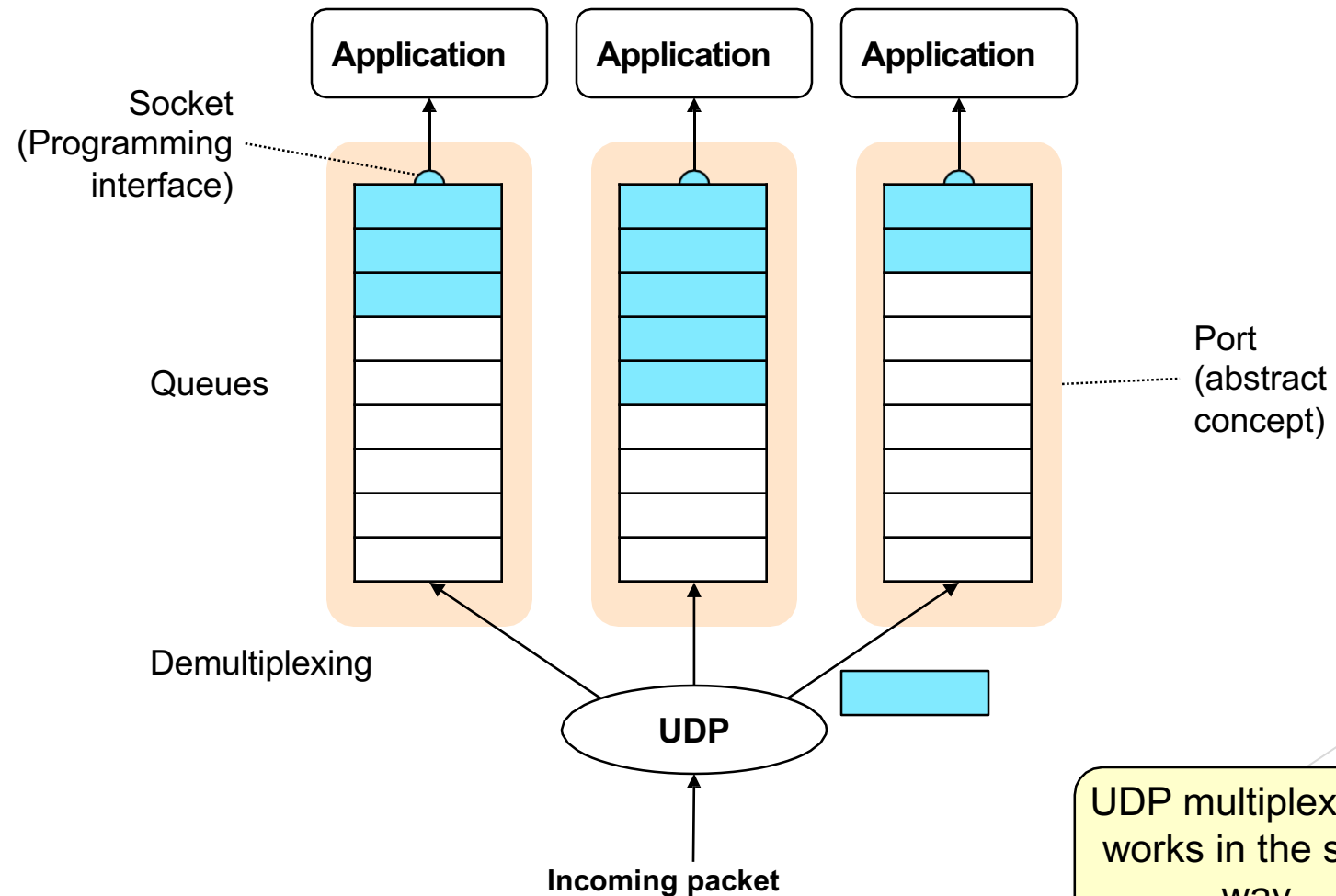
CEN 223 - Internet Communication

Assoc. Prof. Dr. Fatih ABUT
Department of Computer Engineering
Çukurova University

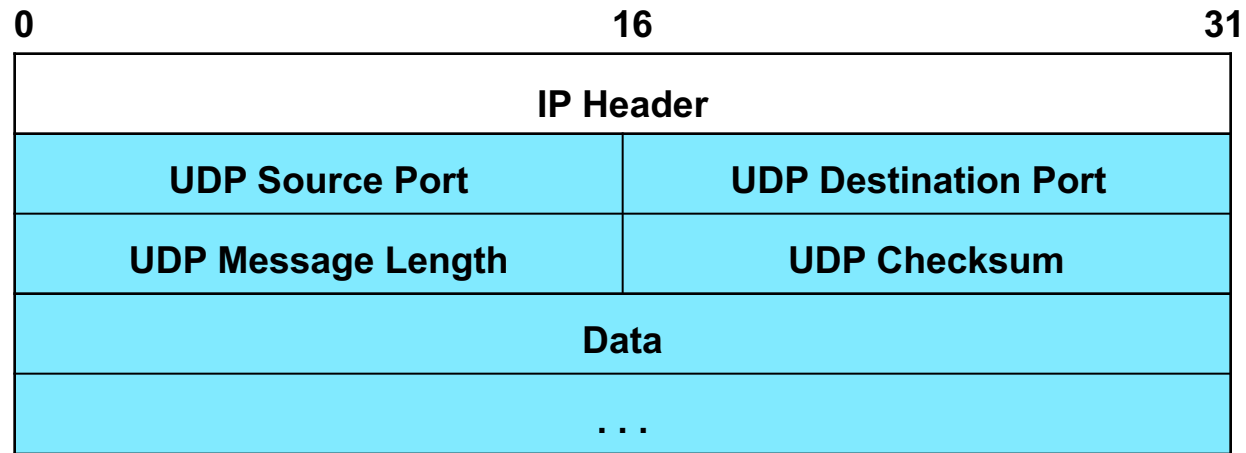
End-to-end protocols: classification + task definition



UDP Demultiplexing (Ports und Sockets)



UDP Header



- Source Port:** Contains the port of the sending host;
Optional specification, i.e., if specified, this is the port to which a possible response should go
- Destination Port:** Destination port where the datagram is sent
- Message Length:** Indicates the total length of the datagram
- Checksum:** Check sum via UDP header and IP pseudo header (protocol number + IP addresses)
Ensures that the correct port and host have been reached.

Problem: Reliable transport performance over an unreliable network layer

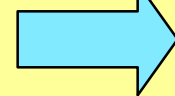
Properties of the IP network layer::

- Dividing the information into packages
- Maximum packet sizes
- Loss of packets in transit
- Receipt of corrupt packets
- Multiple transmission of packets
- Rearranging the order of the packets
- Changing packet transmission times
- Storage capacity of the network

IP is (only) a
packet-oriented
best-effort network

What should TCP do?

**TCP provides a reliable
bidirectional byte stream**



**There is a lot to
be done for TCP**

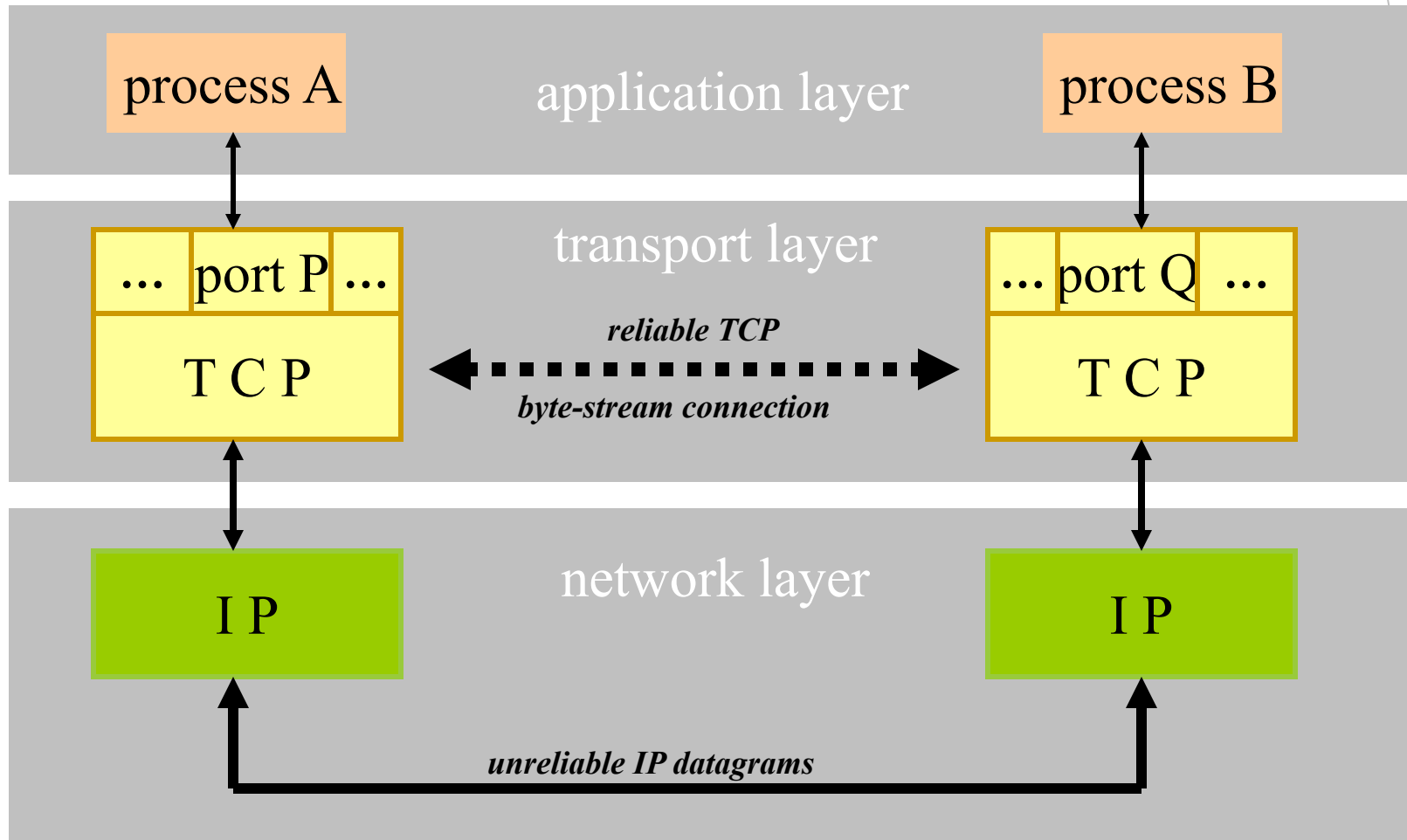
Introduction to TCP

- ▶ Point-to-point connection
- ▶ Streaming interface
 - ▶ Byte-oriented
 - ▶ No boundaries for fragments/segments
- ▶ Reliable, connection-oriented data transfer
 - **Three phases:** connection establishment, user data transmission, connection disconnection
 - Reliable connection establishment (→ **three-way handshake**)
 - Reliable user data transmission (→ **retransmission**)
 - Reliable disconnection
- ▶ Full duplex
- ▶ Flow and congestion controlled
- ▶ Protocol implemented entirely at the ends

Benefits of TCP

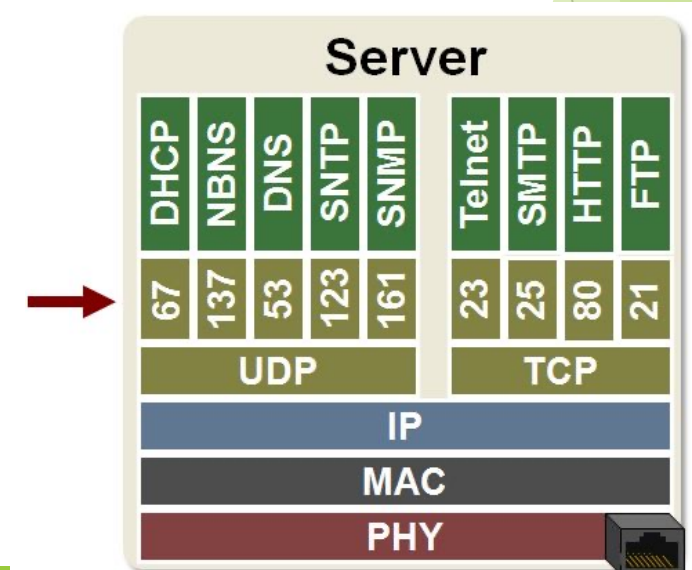
- ▶ Communication is ‘transparently reliable’
- ▶ Data is delivered in the proper sequence
- ▶ An application programmer does not need to worry about issues such as:
 - Lost or delayed packets
 - Timeouts and retransmissions
 - Duplicated packets
 - Packets arriving out-of-sequence
 - Flow and congestion control

Overview of port concept



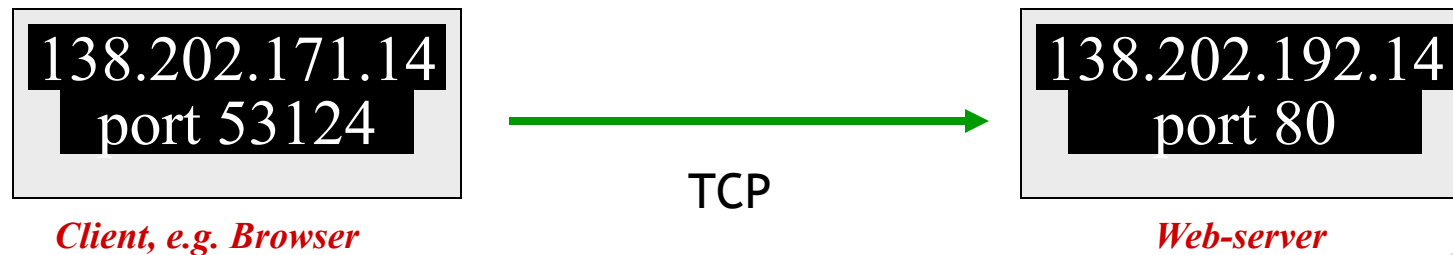
Knowing what port number to use

- ▶ Popular applications have well-known ports
 - ▶ E.g., port 80 for Web and port 25 for e-mail
 - ▶ See <http://www.iana.org/assignments/port-numbers>
- ▶ Well-known vs. ephemeral (temporary) ports
 - ▶ Server has a well-known port (e.g., port 80)
 - ▶ Between 0 and 1023 (requires root to use)
 - ▶ Client picks an unused temporary port
 - ▶ Between 1024 and 65535

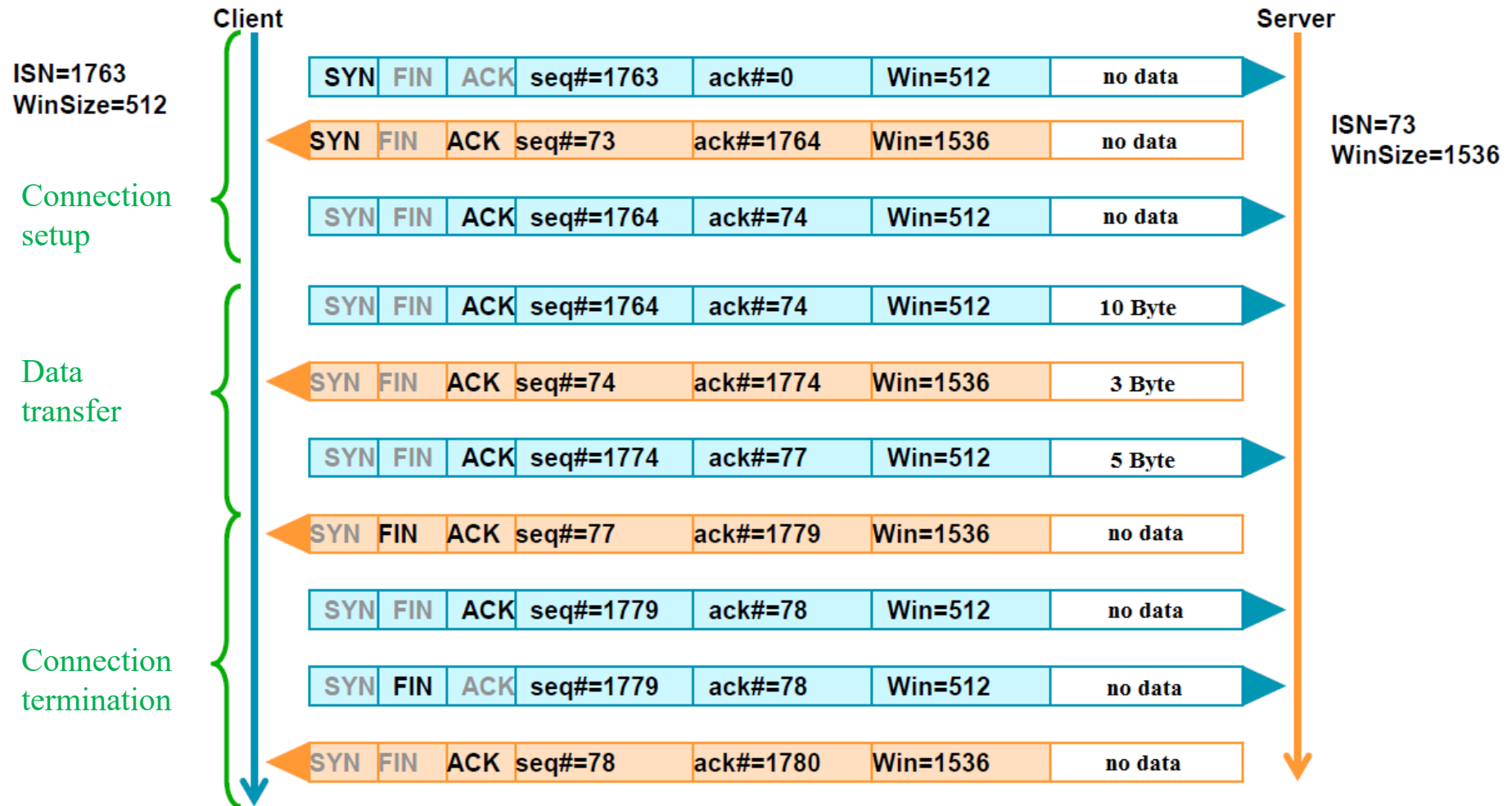


What is a 'connection'?

- ▶ "5 tuple" uniquely identifies traffic between hosts:
 - ▶ An IP-address and port-number for the 'client'
 - ▶ An IP-address and port-number for the 'server'
 - ▶ + underlying transport protocol (e.g., TCP or UDP)



TCP - Normal Procedure (Simplified)



Connection Establishment (cont)

Three way handshake:

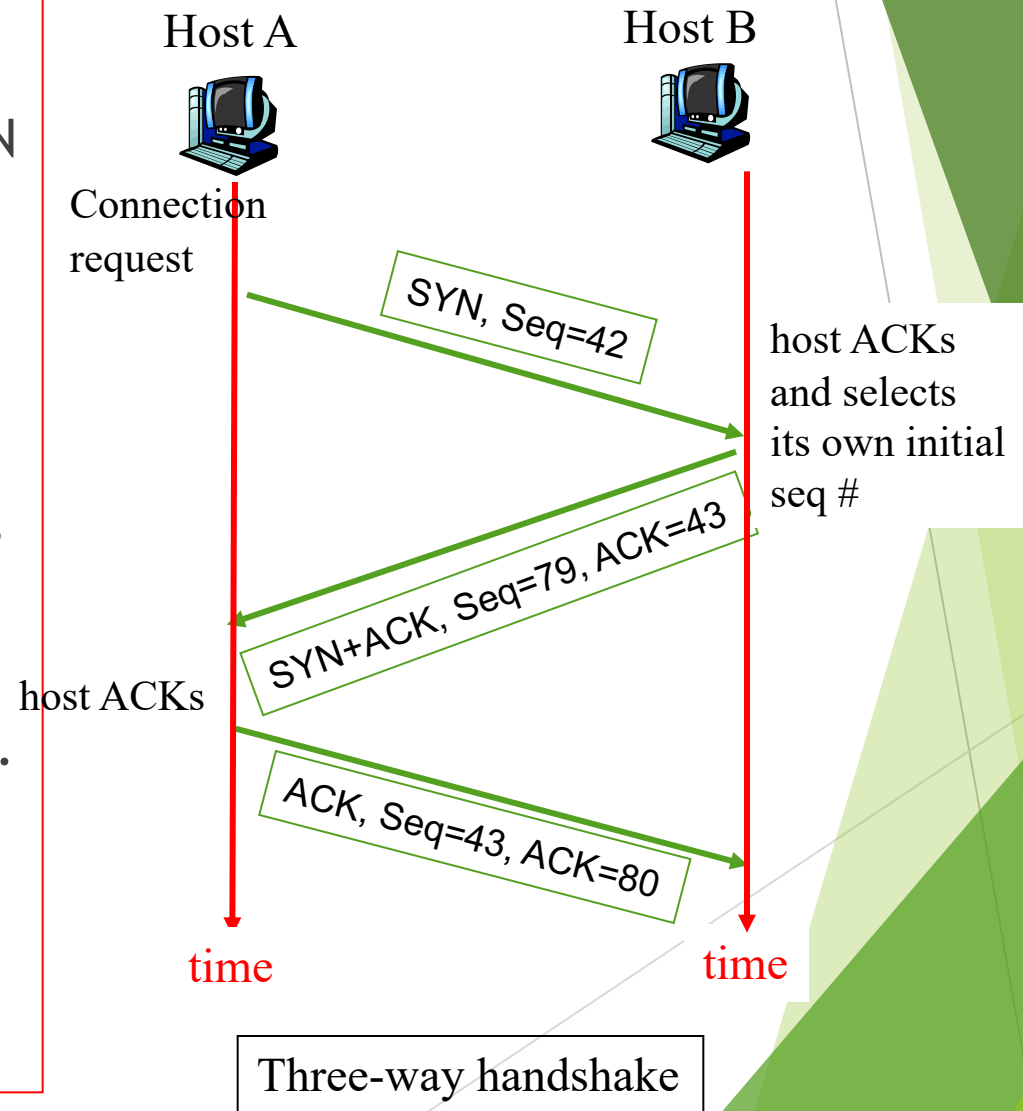
Step 1: client host sends TCP SYN segment to server

- specifies a **random** initial seq #
- no data

Step 2: server host receives SYN, replies with SYNACK segment

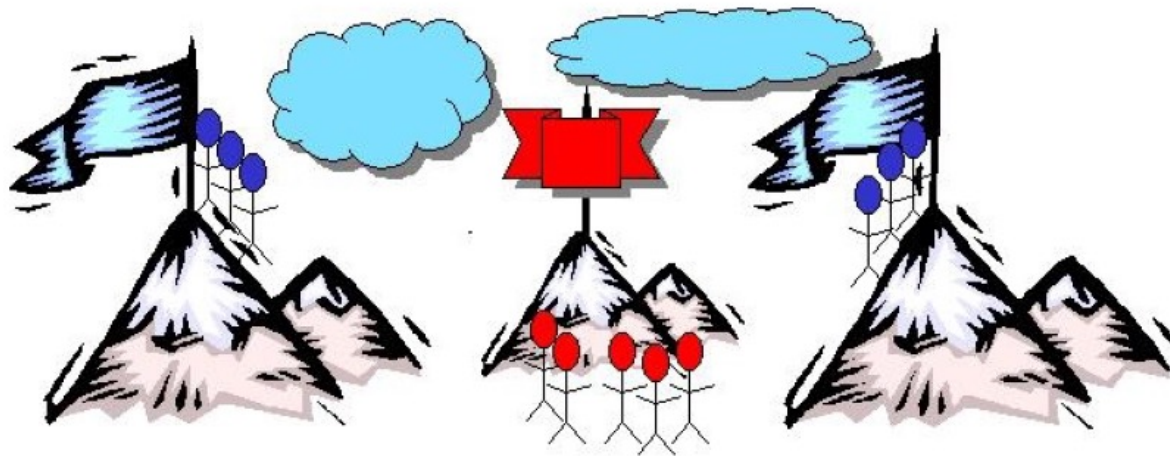
- server allocates buffers
- specifies server initial seq. #

Step 3: client receives SYNACK, replies with ACK segment, which may contain data

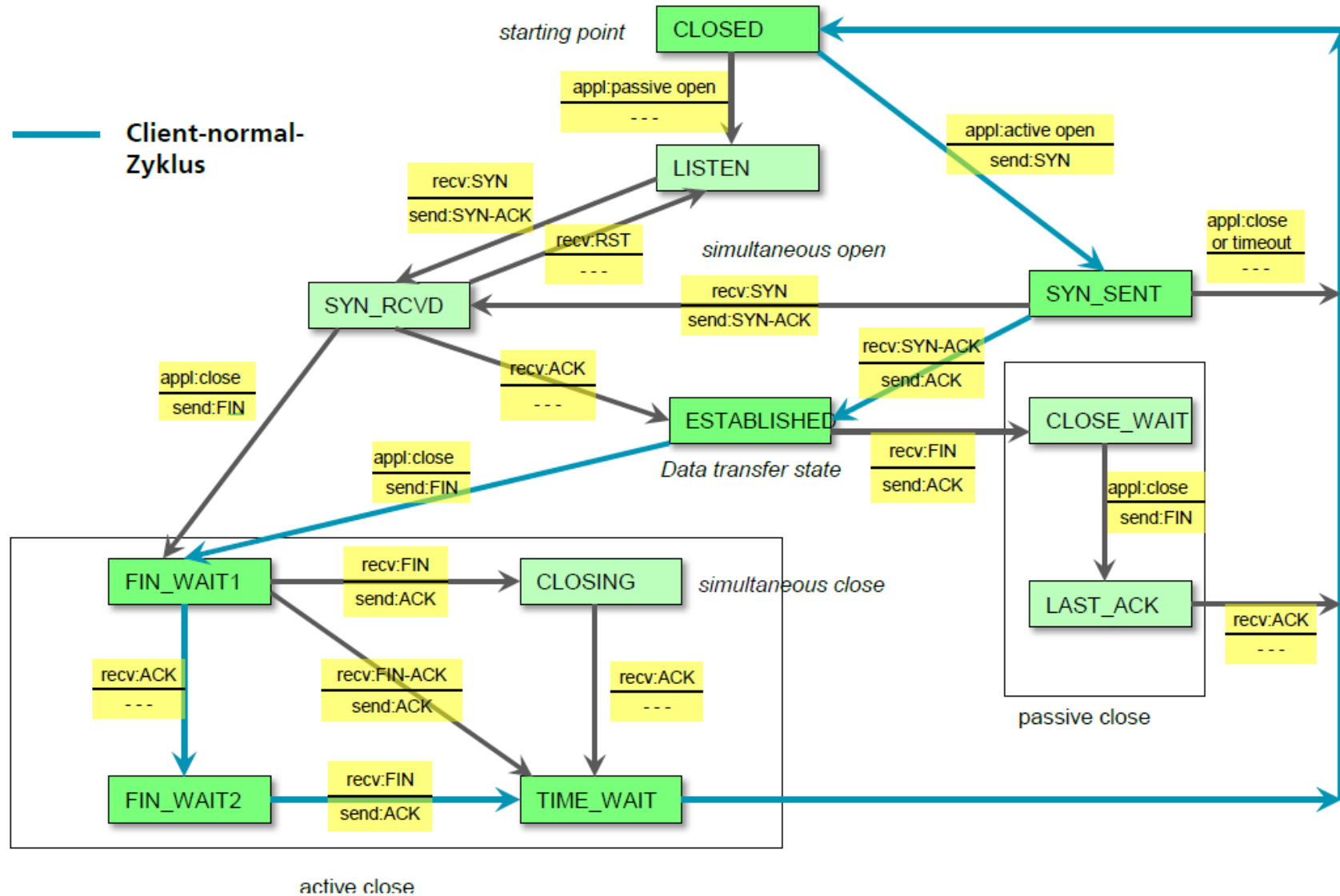


Two-Army Problem

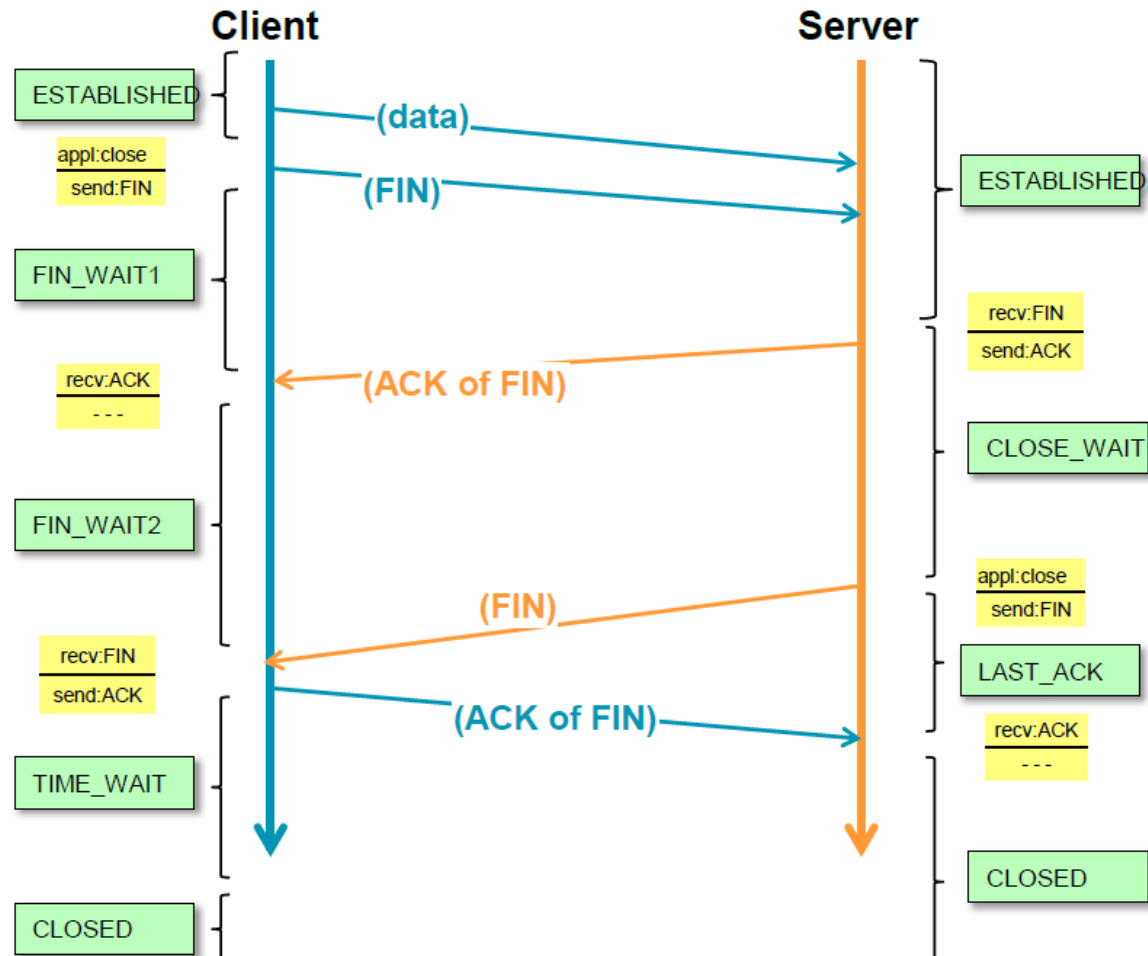
- ▶ Imagine that one army (e.g., red army) is encamped in a valley. On both the surrounding hill sides is opposite army (e.g., blue army). The red army is larger than either of the blue armies alone, but together they are larger than the red army.
- ▶ If either blue army attacks by itself, it will be defeated, but if the blue armies attack simultaneously, they will be victorious.
- ▶ The question is: Does a protocol exist that allows the blue armies to win?



TCP Connection Management



Regular TCP Connection Termination



Terminating a connection - basic types and problem definition

► Basic types

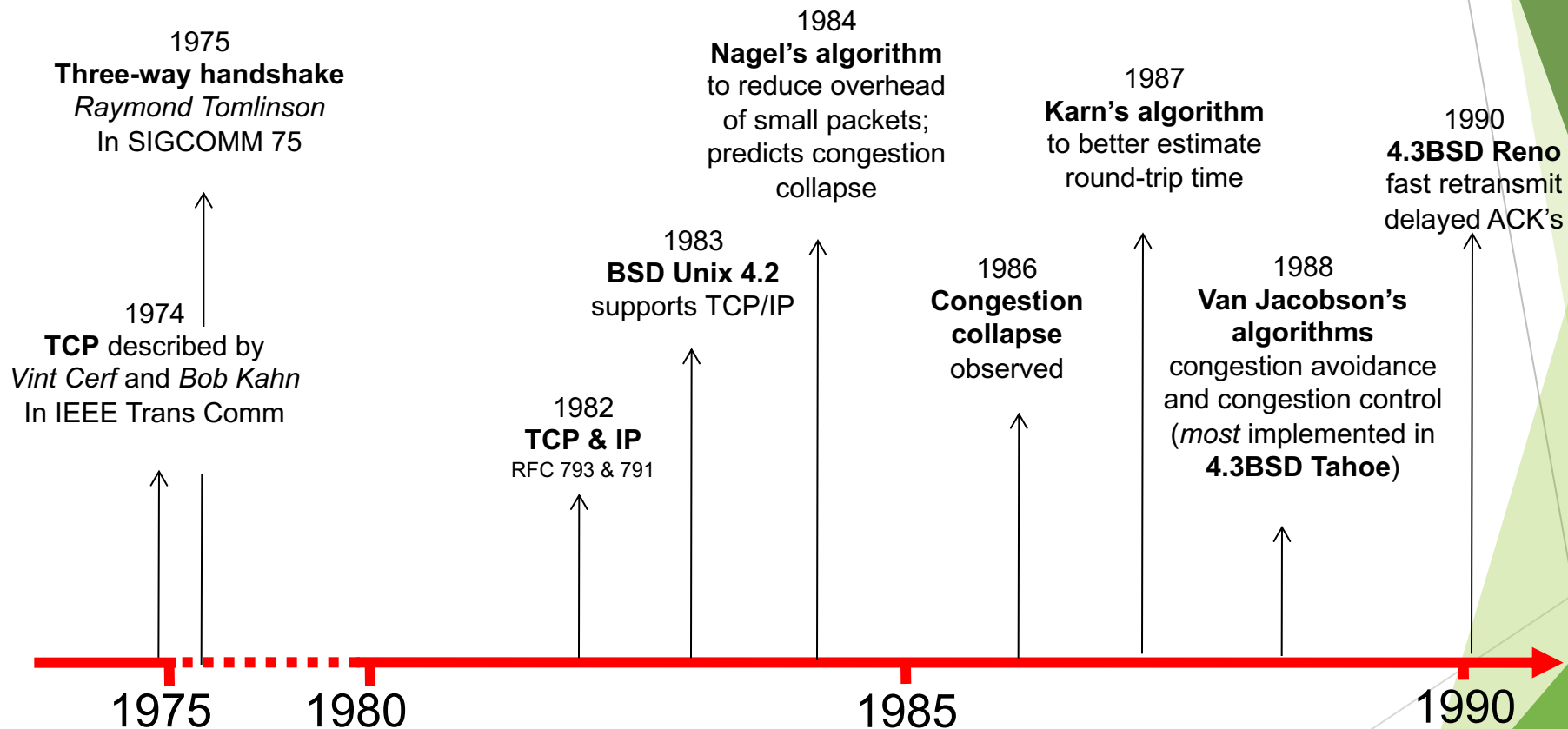
- **Reset:** The connection is terminated by an "asynchronous" side. (Data loss problem)
- **Unilateral closing:** The connection signals to the other side that it is not sending any further data. (Problem point: the other side can send further data)
- **Asymmetrical disconnection:** One of the two computers closes its connection on one side and disconnects the entire connection at the same time.
- **Symmetrical disconnection:** Either of the two sides clears "their" unidirectional connection as soon as both sides have agreed to clear the connection (-> TCP approach)

► Problem with the last ACK: How do I safely terminate a connection?

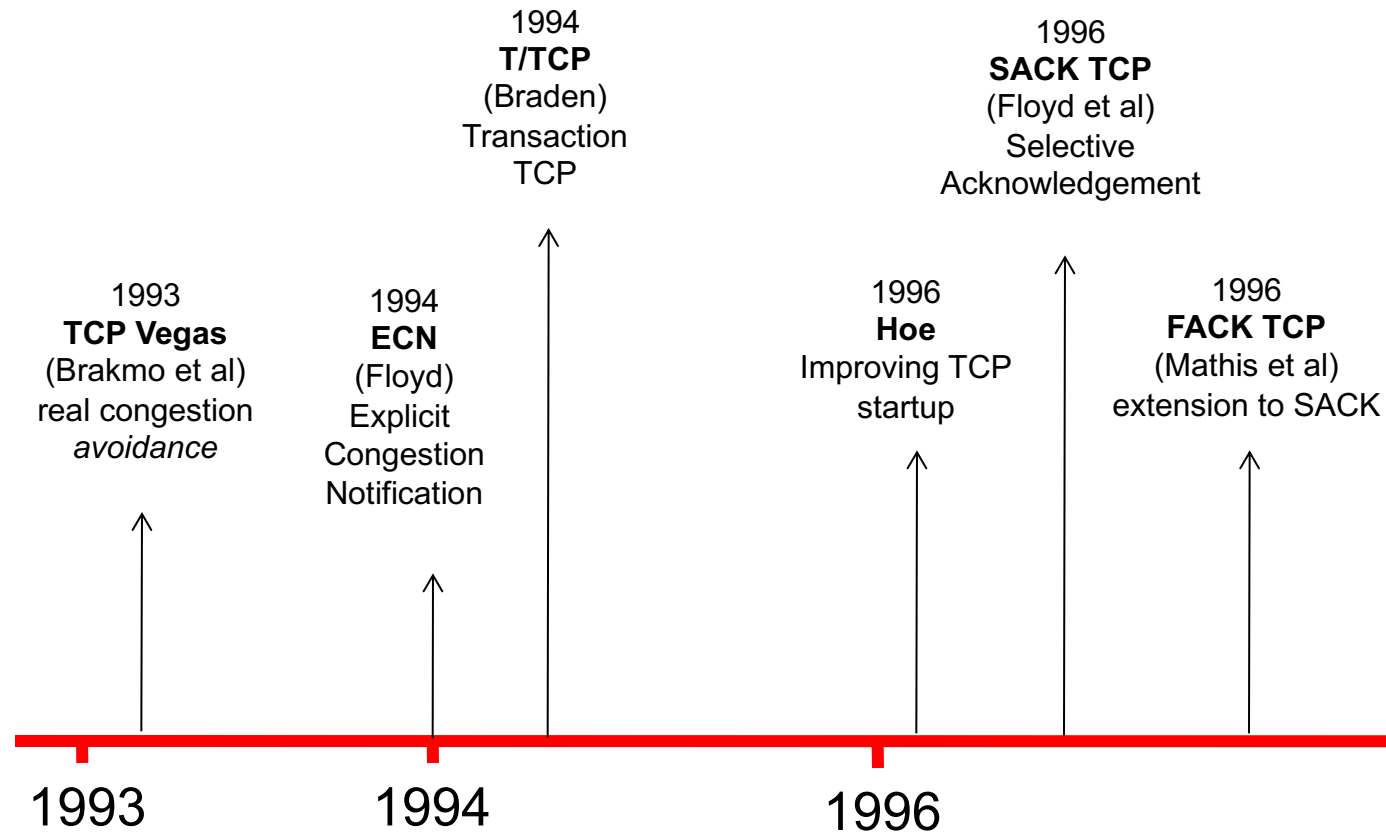
- Is a three-way handshake enough?
- Key question: Is the last PDU important for the protocol?

► Use of timers

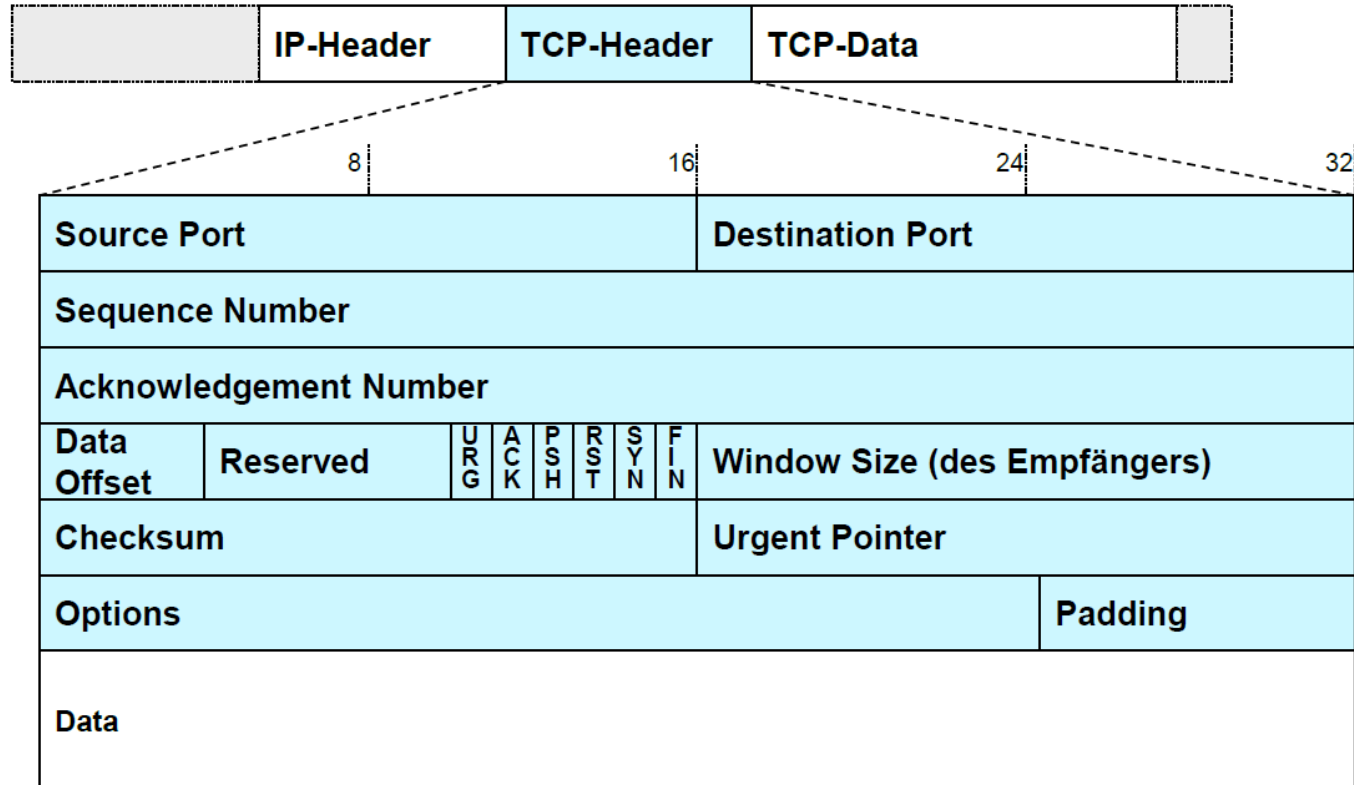
Evolution of TCP



TCP Through the 1990s



TCP Header



Structure of the TCP header

- **Source Port:** 16-bit field for local TCP user (application)
- **Destination Port:** 16-bit field for remote TCP user (application)
- **Sequence number:** Position of the current block in the current stream. Points to the first new byte.
- **Acknowledgment Number:** Next sequence number expected. Previous numbers were received correctly.
- **Data Offset:** Length of the TCP header in 32-bit blocks
- **Flags:**
 - **URG** Urgent-Pointer-Flag (for priority data)
 - **ACK** Acknowledgement-Flag (confirmation)
 - **PSH** Push-Flag (direct forwarding to application)
 - **RST** Reset-Flag (Termination of the connection due to an error)
 - **SYN** Synchronization-Flag (Synchronization process)
 - **FIN** Final-Flag (proper disconnection)

Structure of the TCP header (Cont.)

- **Window Size:** The buffer size still available on the receiving side, important information for the transmitter, is used for flow control
- **Urgent Pointer:** Pointer to the end of urgent data
- **Cchecksum:** Check sum of headers, data and a pseudo header to protect against misdirected segments
- **Options:** MSS (Maximum Segment Size) when establishing a connection
- **Padding:** Fill data for 32 bit limit